Assimilation Eye — User Guide

How to power, connect to, configure and troubleshoot the prop. Nothing here needs a PC or any programming: a phone is enough.

- To build the circuit, see hardware-guide.md.
- To build or change the firmware, see README.md.

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1. Quick reference

Factory settings. All of them can be changed, and a factory reset restores them.

Prop's own Wi-Fi network	`AssimilationEye`
Password for that network	`resistanceisfutile`
Address on that network	`http://192.168.4.1`
Sign-in user name	`admin`
Sign-in password	`assimilate`
Name on a joined network	`http://assimilation-eye.local`
Factory reset	Hold the BOOT button for 10 seconds

The address always starts with `http://`, not `https://`.

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2. First power-on

1. Plug the board into USB power: a power bank, a phone charger or a PC.
2. The board's red power light comes on.
3. **Self-test.** LEDs 1 to 8 light one after another, clockwise as the audience sees the ring, then the whole ring fades up and down once. This takes about three seconds.

4. The ring starts its pattern. A new or factory-reset prop starts on **Comet and flash**.
5. About five seconds after power-on the prop's own Wi-Fi network, `AssimilationEye`, is available.

If the self-test skips an LED or runs out of order, see Troubleshooting.

The prop needs no phone to run. It starts the pattern and settings it was last left with.

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3. Connecting and signing in

There are two ways to reach the prop. Both show the same screens and take the same sign-in.

3.1 Through the prop's own network (always works)

1. On the phone, open the Wi-Fi settings and join `AssimilationEye` with the password `resistanceisfutile`.
2. The phone will say the network has no internet. That is correct. If it asks whether to stay connected, choose to stay.
3. Open a browser and go to `http://192.168.4.1`.

If the page does not load, turn off mobile data for a moment. Some phones send everything over mobile data when Wi-Fi has no internet.

3.2 Through a network the prop has joined

Once the prop has joined a Wi-Fi network (section 5), any phone or PC on that network can open it at:

- the address shown on the Wi-Fi screen under **Joined network**, for example `http://192.168.1.50`, or
- `http://assimilation-eye.local`. This name works on iPhones, Macs and recent Windows. Many Android phones do not support it; use the address.

The prop's own network stays on as well, so section 3.1 always remains available.

3.3 Signing in

Enter the user name and password. On a new prop these are `admin` and `assimilate`.

- A banner reminds you while the password is still the factory default. Change it on the System screen (section 6.3).
- After **five wrong attempts** sign-in is locked for **30 seconds**. The screen counts down.
- A sign-in lasts until the prop restarts, until you sign out, or 12 hours without use.

3.4 Finding your way around

Three tabs across the top: **Eye**, **Wi-Fi** and **System**.

The pill at the top right shows the link to the prop:

Pill	Meaning
Live	Changes take effect at once

Pill	Meaning
Connecting…	The page has just opened
Reconnecting…	The page has lost the prop and is trying again. The LEDs carry on regardless. Controls are greyed out until it is back

Changes apply immediately; there is no Save button on the Eye screen. The prop stores a change about two seconds after you stop adjusting.

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4. The Eye screen

4.1 Ring

Control	What it does
Stop the ring / Start the ring	Turns the ring off and on. The setting is remembered
Pattern	Rotating comet, Synchronized flash, or Comet and flash
Overall brightness	Scales everything, 0–100%

The cards below change with the pattern chosen.

4.2 Rotating comet

One bright LED travels around the ring, followed by a fading tail.

Control	Range	What it does
Direction	Clockwise / Counter-clockwise	As the audience sees the ring
Speed	20 ms to 2 s per LED	Time the head spends on each LED. The hint shows the time for one lap
Head brightness	0–100%	The leading LED
Tail length	0–7 LEDs	LEDs behind the head. 4 gives five lit LEDs in all
Tail fade	10–90% dimmer per LED	How quickly the tail dies away. Higher is a shorter-looking tail

The head glides from one LED to the next, so even slow speeds look smooth.

4.3 Synchronized flash

All eight LEDs pulse together.

Control	Range	What it does
Interval	0.1–10 s	From the start of one flash to the start of the next

Control	Range	What it does
Ramp up	Abrupt to 5 s	Time to rise from minimum to maximum
Hold at maximum	0–10 s	Time spent at full
Ramp down	Abrupt to 5 s	Time to fall back
Maximum brightness	0–100%	The top of the pulse
Minimum brightness	0–100%	The level between pulses. 0 is dark

Ramp up, hold and ramp down must fit inside the interval. If they do not, the prop shortens the hold first, then the ramps, and the sliders move to show what it settled on. The minimum can never be above the maximum.

For a hard strobe, set both ramps to Abrupt. For a slow breathing glow, use long ramps and a short hold.

4.4 Comet and flash

Plays the comet, crossfades into the flash, and crossfades back, for as long as it runs. It uses the comet and the flash exactly as you have set them, and both cards stay on screen so they can be adjusted while the combined pattern plays.

Control	Range	What it does
Time on comet	1–60 s	How long the comet plays each round
Time on flash	1–60 s	How long the flash plays each round
Crossfade	Cut to 5 s	How long each blend between the two takes
Comet direction	Random / As set for the comet	See below

With **Random**, the comet's direction is picked afresh each time it returns. The choice is made while the flash is showing, so the comet never visibly reverses. The Direction control on the Comet card is hidden while Random is on. With **As set for the comet**, the comet always turns the way chosen on the Comet card.

4.5 Onboard LED

The small blue LED on the board itself. It is separate from the ring. A board without one shows a note instead of these controls.

Behaviour	What it does
Off	Stays dark
On	Stays lit
Heartbeat	A double beat, repeating, to show the prop is alive
Blink on change	Blinks when the ring pattern is switched. Blinks and Blink length set how many and how long

4.6 Reset LED settings

At the bottom of the screen. Tap once, then tap again to confirm. It restores every setting on this screen to the defaults this firmware was built with. Sign-in and Wi-Fi are not touched.

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5. The Wi-Fi screen

5.1 Joining a network

1. Tap **Search for networks**. After a few seconds a list appears with signal strength.
2. Tap the network. Its name is filled in. A network that is hidden can be typed in instead, exactly as it is spelt.
3. Enter the password. Leave it empty for an open network.
4. Leave **Address** on **Automatic (DHCP)** unless you need a fixed address.
5. Tap **Save and join**.

Within about 20 seconds the **Joined network** card at the top shows the result. When joined, it shows the address to use from that network.

If you are connected through the prop's own network while doing this, the page may lose contact for a moment while the prop changes channel. It reconnects by itself.

The prop only sees **2.4 GHz** networks. A network that asks for a user name as well as a password, or shows a sign-in web page (common in hotels and offices), cannot be joined.

5.2 A fixed (static) address

Choose **Static** under Address and fill in:

Field	Example	Notes
IP address	`192.168.1.50`	Pick one outside the range the router hands out by itself
Subnet mask	`255.255.255.0`	
Gateway	`192.168.1.1`	Usually the router's address. Must be in the same subnet as the IP address
DNS server		Optional. Empty means use the gateway
Second DNS server		Optional

Settings that cannot work are refused with a message saying why. Addresses in `192.168.4.x` are not allowed, because the prop's own network uses them.

A static address that is valid but wrong for the network, for example one already used by another device, cannot be detected. If the prop becomes unreachable there, connect through its own network (section 3.1) and correct it.

To go back, choose **Automatic (DHCP)** and tap **Save and join**. For the network already saved, the password can be left empty to keep it.

5.3 If the network is unavailable

A wrong password, or a saved network that is out of range, does no harm. The card says it could not join and why, the prop keeps trying every half minute, and it stays reachable on its own network throughout.

Forget this network removes the saved network.

5.4 The prop's own access point

The last card changes the name and password of the prop's own network. The password must be 8 to 63 characters. Saving disconnects every device on that network; join again with the new details.

A warning shows while the factory password is still in use.

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6. The System screen

6.1 This eye

Firmware version and build time, whether the LED defaults were set for this build, running time, the reason for the last start, memory and firmware size.

A warning appears on a development build. Such a build has a Wi-Fi name and password compiled in and must not be passed on.

6.2 Update firmware

1. Get the `firmware.bin` file from whoever built the firmware.
2. Tap the file chooser and select it.
3. Tap **Install this firmware**, then tap again to confirm.
4. Keep the page open. The ring pulses slowly while the firmware is installed.
5. The prop restarts and the sign-in screen returns. The self-test runs again.

LED settings, sign-in details and the saved network are all kept.

A file that is not firmware for this board, or is too large, is refused with a message, and the prop carries on with the firmware it has. An upload that is interrupted also leaves the running firmware in place; start again.

6.3 Sign-in details

Enter the current password, then the user name (3 to 32 characters) and the new password twice (8 to 64 characters). Saving signs out every other browser.

If the new details are lost, a factory reset (section 7) restores `admin` and `assimilate`.

6.4 Signed in

Sign out ends the session in this browser.

6.5 Restart and reset

- **Restart the eye** restarts it. Nothing is lost.
- **Factory reset** needs the current password. It does the same as holding the BOOT button, described next.

7. Factory reset with the BOOT button

Use this when the sign-in details are lost or the prop is unreachable.

1. With the prop powered and running, press and hold the **BOOT** button on the board.
2. After 3 seconds the whole ring starts to blink as a warning. Letting go now cancels.
3. Keep holding. At 10 seconds the ring lights steadily for two seconds. Let go.
4. The prop restarts.

What it restores:

	Restored to
Sign-in	`admin` / `assimilate`
Saved Wi-Fi network	Forgotten
Prop's own network	`AssimilationEye` / `resistanceisfutile`
LED settings	The defaults this firmware was built with

The firmware itself is not changed.

Do not hold BOOT while plugging in the power. That puts the board into its programming mode: the ring stays dark and nothing runs. Unplug it and plug it in again without touching the button.

8. At the demo venue

Use the prop's own network. It needs nothing from the venue, it cannot be blocked by the venue's Wi-Fi rules, and no one else is on it.

1. Power the prop.
2. Join `AssimilationEye` from the phone that will control it, and open `http://192.168.4.1`.
3. Sign in, set the pattern, and leave the page open.

Practical points:

- **Range** is a room, not a building. Keep the controlling phone within about ten metres.
- **Stop the phone leaving the network.** Tell it to stay connected when it warns there is no internet, and turn off any "switch to mobile data" option for the duration.
- **The prop does not need the phone.** If the phone drops off, the ring carries on with the last settings. Reconnect when convenient.
- **Switching pattern mid-demo** is one tap on the Eye screen. For a dramatic start, tap **Stop the ring** beforehand, put the prop on, then tap **Start the ring**.
- **Before unplugging** after changing a setting, wait two seconds so the change is stored.
- **Do not use the venue's Wi-Fi for control** unless you have tested it there. Many venue networks stop devices from talking to each other.

A word on security. The connection is not encrypted. On a shared network, someone able to capture traffic could read the password as you sign in. On the prop's own network, with its password changed from the

factory one, only people you have given that password to can connect. Change both passwords before a public demo.

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9. Pre-demo checklist

The day before:

- ☐ Power bank fully charged, and tested with the prop for at least 15 minutes without cutting out
- ☐ Sign-in password changed from the factory default
- ☐ Access point password changed from the factory default, and the new one saved on the controlling phone
- ☐ Pattern and settings chosen, and the prop switched off and on once to confirm it comes back to them
- ☐ All eight LEDs light in the self-test, in order
- ☐ Wiring and joints secure; nothing pulls when the cap or glasses move

On the day:

- ☐ Phone joined to the prop's network and the page open at `http://192.168.4.1`
- ☐ Pill at the top right reads **Live**
- ☐ Phone set not to sleep or switch network during the demo
- ☐ Ring stopped or running, as the script needs
- ☐ The off switch is within reach

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10. Troubleshooting

10.1 Connecting, signing in and updating

Symptom	Likely cause	Fix
`AssimilationEye` is not in the phone's Wi-Fi list	The prop is not powered or has not finished starting	Check the board's red power light. Wait ten seconds after power-on and refresh the list
	The network name was changed	Look for the new name, or factory reset (section 7)
	The board is in programming mode because BOOT was held at power-on	Unplug and plug in again without touching BOOT
The phone will not join `AssimilationEye`	Wrong password	The factory password is `resistanceisfutile`. If it was changed and lost, factory reset
	The phone remembers an old password	Forget the network on the phone and join again

Symptom	Likely cause	Fix
Joined, but `192.168.4.1` does not load	The phone is using mobile data because the network has no internet	Turn off mobile data, or tell the phone to stay on this network
	The browser changed the address to `https://`	Type `http://192.168.4.1` in full
	A VPN or private relay is on	Turn it off for this network
The page loads slowly or hangs once, then is fine	The board occasionally misses a new connection	Reload the page
`assimilation-eye.local` does not open	The phone or PC does not support `.local` names (many Android phones)	Use the numeric address from the Wi-Fi screen
	The phone is not on the same network as the prop	Join the same network, or use the prop's own
The address that worked yesterday does not work	The router gave the prop a different address	Read the new address from the Wi-Fi screen through the prop's own network. For an address that never changes, set a static one (section 5.2)
"Too many wrong attempts"	Five wrong sign-ins	Wait 30 seconds
The user name or password is forgotten	—	Factory reset (section 7)
Signed out unexpectedly	The prop restarted, or the sign-in details were changed from another browser	Sign in again
Pill shows **Reconnecting…**	The phone left the network or moved out of range	Rejoin the network. The ring is still running
The prop will not join a network	Wrong password	Enter it again. The card says "the network refused the password" when this is the cause
	The network is 5 GHz only	Use a 2.4 GHz network, or a phone hotspot set to 2.4 GHz
	The network needs a user name or a sign-in web page	It cannot be joined. Use the prop's own network

Symptom	Likely cause	Fix
Joined, but the prop cannot be reached at its address	The network stops devices talking to each other	Use the prop's own network
	A static address clashes or is in the wrong range	Correct it through the prop's own network, or switch back to DHCP
Firmware upload is refused	The file is not a firmware for this board, or it is too large	Use the `firmware.bin` from the build. The prop is unharmed
Firmware upload stops part-way	The connection dropped	The previous firmware is still running. Move closer and upload again
A setting did not survive switching off	Power was removed within two seconds of the change	Make the change again and wait before unplugging
The ring hesitates for an instant about two seconds after a change	The setting is being stored	Normal. It does not happen while nothing is being changed

10.2 Hardware

Symptom	Likely cause	Fix
Nothing at all: no red power light	No power reaching the board	Try another cable, another port or another power bank. Check the inline switch
	The cable is damaged	Replace it
Red power light is on, the ring is dark, no Wi-Fi network	The board is in programming mode because BOOT was pressed at power-on	Unplug and plug in again without touching BOOT. On the wearable, check nothing is pressing on the button
Red light on, Wi-Fi network present, the ring is dark	The ring is stopped, or brightness is at 0	On the Eye screen tap **Start the ring** and raise **Overall brightness**
	The ground wire from the board to the LEDs has come off	Reconnect `GND` to the common cathode wire
One LED never lights, including in the self-test	The LED is the wrong way round	Turn it: long leg to the resistor, short leg to ground
	A loose wire, resistor leg or solder joint	Reseat or re-solder. Move the wiring gently while watching the LED
	The LED is dead	Swap it with a neighbour. If the fault moves with the LED, replace it

Symptom	Likely cause	Fix
The self-test lights the LEDs out of order	Two signal wires are swapped	Compare against the table in section 7 of the hardware guide and swap them back
The comet runs the wrong way round for the audience	The ring was assembled in mirror image	Choose the other **Direction** on the Eye screen. Nothing needs rewiring
All LEDs are dim	Brightness settings are low	Raise **Overall brightness**, and **Head brightness** or **Maximum brightness**
	220 Ω is not bright enough under stage lighting	Change all eight resistors to 100 Ω (hardware guide, section 3)
One LED is dimmer than the rest	A resistor of the wrong value	Check the colour bands: red, red, brown for 220 Ω
	A poor joint adding resistance	Re-solder it
An LED flickers or cuts out when the prop moves	A loose connection or a wire breaking at a joint	Re-solder, and add strain relief so the wire does not flex at the joint
Two LEDs always light together	Their signal wires or legs are touching	Separate and insulate them
The ring looks striped or flickers on camera	The camera's shutter is beating with the LED dimming	Raise the brightness, which reduces the effect, or change the camera's shutter speed
The prop switches off after half a minute to a few minutes on a power bank	The power bank shuts down because the prop draws too little current for it to notice	Use a power bank with a low-current or "always on" mode, or a different power bank. Test for 15 minutes before the demo
The prop restarts by itself	A loose USB plug, or a weak power bank	Reseat the plug; try another cable or bank. The System screen shows the reason for the last start
The board is hot, or the PC reports a USB fault	A short circuit	Unplug at once. Look for bare wires touching, especially between `3V3` and `GND`
The PC does not detect the board	A charge-only USB cable	Use a data cable
	The driver for the board's USB chip (CP210x) is missing	Install the CP210x driver from Silicon Labs
Uploading from the PC fails with "Failed to connect"	The board did not enter programming mode by itself	Hold BOOT while the upload starts, and let go when it begins writing

Symptom	Likely cause	Fix
The blue onboard LED never lights	Its behaviour is set to Off	Choose another behaviour on the Eye screen
	The board has none (genuine 38-pin DevKitC)	Nothing to fix. The Eye screen says so

10.3 When nothing else works

1. Unplug the prop, wait five seconds, plug it in again without touching any button.
2. Watch the self-test. If it runs, the board, firmware and LEDs are fine and the problem is in connecting.
3. Connect through the prop's own network (section 3.1).
4. If you cannot sign in, factory reset (section 7).
5. If there is no self-test and no Wi-Fi network, the firmware needs to be put back from a PC over USB. See README.md.